

I

PART

Introduction to Skill Acquisition Theory and the Constraints-Led Approach

Part 1 provides an overview of the key theoretical contributions to the study of skill acquisition and introduces the constraints-led approach.

Chapter 1 reviews the contribution of traditional theories to the study of skill acquisition, including the information-processing approach. In this chapter, we discuss important practical considerations such as practice organization and feedback delivery to provide a basis for comparison in later chapters. In chapters 2 and 3, we explore the relevance to motor performance of some alternative theoretical insights, particularly those offered by Russian physiologist Nikolai Bernstein (1967) and American psychologist James Gibson (1979). By emphasizing the need to adopt a systems perspective of human behavior, their theoretical contributions have had a major impact on

our understanding of movement coordination and control. Their insights have encouraged researchers to model the performer as a complex movement system composed of many interacting subsystems. These ideas have focused attention on how coordination emerges between the parts of each person's movement system, as well as the key variables or constraints that the person uses to regulate or guide these coordination patterns.

Clearly, constraints play an important role in shaping the ways in which humans can move to effect change in their environment. In particular, chapter 2 focuses on the physical constraints affecting the performer by discussing key concepts of dynamical systems theory, and chapter 3 describes ecological psychology as a suitable framework for understanding informational constraints on behavior. In chapter 4, we present an overview of the constraints-led approach to skill acquisition, which forms the foundation of part II. Although there are a number of models of constraints, in part I we focus on the implications of Newell's (1986) model as a template for understanding how motor skills are acquired.

Traditional Theories of Skill Acquisition

CHAPTER OUTLINE

Skill Acquisition: Definition and Theories

- Association Theories
- Neuromaturational Theories
- Fitts' Stage Theory of Motor Learning
- Information-Processing Theories
- Neurocomputational Theories

Common Features of Traditional Theories

- Representations of the World and Movements
- Learner as Hierarchical Control System
- Capacity-Limited Storage
- Indirect Perception
- Reduction of Error and Noise
- Models of Limited Range of Movement

Implications for Movement Practitioners

- Amount of Practice
- Practice Variability
- Practice Organization
- Part-Task Practice
- Instructions and Feedback
- Demonstrations of Technique

Summary

Over the last century, the question of how humans learn to control and coordinate their movements has received much attention, with scientists proposing many different theories. Although there is still considerable debate over which theory is most appropriate (Summers, 2004), the importance of developing a strong theoretical framework for studying skill acquisition and guiding practical activity remains clear. Many practitioners use models of human behavior, either implicitly or explicitly, to plan their decision making for focused, effective practical activity. Theories of skill acquisition can help practitioners develop an appropriate model of the motor learning process that is necessary for understanding how learners acquire motor skills. How do you currently view motor behavior and how does your understanding underpin the way you teach or coach? As part II of this book will show, this knowledge provides a philosophy for structuring practice contexts, providing information to learners, using visual demonstrations, and other important tasks.

This chapter provides a brief historical overview of the traditional theories of movement skill acquisition. This overview will enable us to compile a number of implications for movement practitioners. Summarizing the general features of traditional theories will help us compare them with alternative theories described in subsequent chapters. As you read this chapter, we encourage you to keep an open but critical mind regarding the strengths and weaknesses of the different theories discussed. (For more detailed overviews of these theories, refer to Schmidt and Wrisberg [2004], Magill [2006], and Haywood and Getchell [2005]). Let's start by considering what we mean by the term *skill acquisition*.

Skill Acquisition: Definition and Theories

Motor skill acquisition has traditionally been described as the internal processes that bring about relatively permanent changes in the learner's movement capabilities (Schmidt & Wrisberg, 2004). For example, movement skills such as riding a bicycle, catching a ball, or driving a car require a good deal of practice to allow us to perform them effectively. **Skill acquisition** requires us to interact effectively with our environment, detect important information, and time our responses appropriately. It thus should result in coordination patterns that are adaptable to a range of varying performance characteristics. Adaptive behavior is important because conditions like the environment, task requirements, and our motivations can change every time we perform a motor skill (Davids, Bennett, & Newell, 2006).

The process of skill acquisition is distinct from execution of the skill (motor control) in that learning is a gradual process that occurs over many performance attempts, resulting in behavior that is less vulnerable to transitory factors such as fatigue, audience effects, and anxiety. One way researchers have tried to understand skill acquisition is by examining performance changes over time. For example, in early research efforts, Bryan and Harter (1897) studied

how learners' typing skills developed while practicing to send and translate Morse code. Over a period of 40 weeks, the telegraphers went through distinct phases of improvement and periods where performance levels plateaued. The inference from such performance curves was that learners initially construct simple elements of the skill, interspersed with periods of consolidation (i.e., development of automaticity), before linking individual parts of movements (in this case, individual finger presses) into more integrated patterns of behavior (linked sequences of finger presses typed as words).

Snoddy (1926) proposed that learning can be described in a predictable manner using a mathematical equation ($\log C = \log B + n \log x$, where C is a measure of performance, x is the amount of practice, and B and n are constants). This **power law of practice** predicts a straight-line relationship between the logarithmic functions of practice time and performance (see figure 1.1). The power law of practice simply states that performance improves with practice, although there are eventual physical limits to this relationship.



Practicing a skill such as riding a bicycle leads to performance improvement. Skill acquisition helps us to interact effectively with our environment, detect important information, and time our responses appropriately.

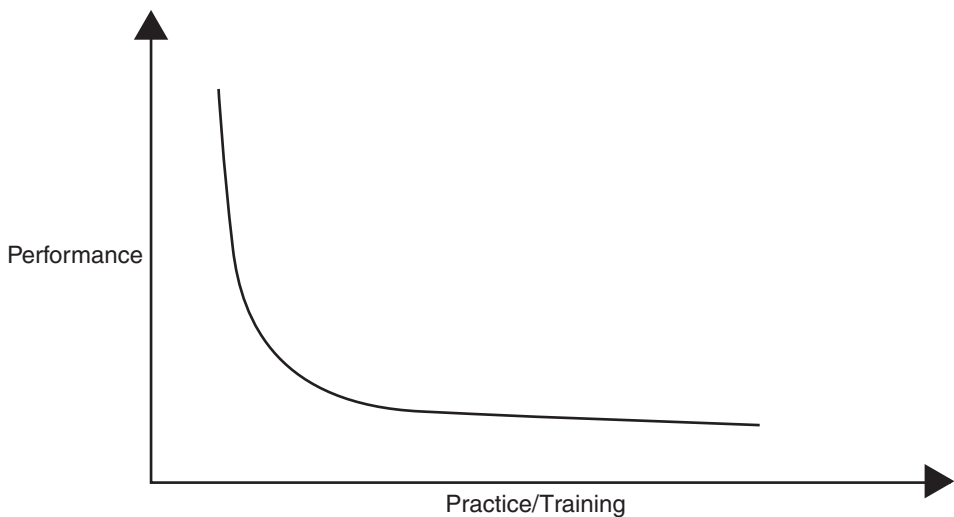


Figure 1.1 The power law of practice predicts a straight-line relationship between the logarithmic functions of practice time and performance.

Despite being researched for some years, the power law of practice has failed to gain universal acceptance because it has become clear that learning is often characterized by sudden jumps, rapid improvements, and even decrements in performance over time. For example, although Crossman's (1959) classic study of cigar makers seemed to support Snoddy's power law, researchers continue to question the generality of this prediction because many types of learning curves have been observed in motor learning experiments as well as practice (see Newell, Liu, & Mayer-Kress, 2001).

Initially, researchers favored a much broader definition of skill acquisition that included movement activities with a heavy emphasis on cognition, such as playing chess or learning new languages (e.g., Bryan & Harter, 1897; Chase & Simon, 1973; Ebbinghaus, 1964). Indeed, research examining such skills has contributed significantly to our understanding of processes such as expertise, attention, and automaticity of movement control. More recently, there has been a tendency to examine the acquisition of movement coordination and control with an equal theoretical emphasis on cognition, perception, and action. Before we discuss this more recent trend in later chapters, we will briefly describe five traditional theoretical approaches to skill acquisition: association theories, neuromaturational theories, Fitts' stage theory of motor learning, information-processing theories, and neurocomputational theories.

Association Theories

Early contributors to the study of skill acquisition were interested in the relationship between movement stimuli (information) and action (e.g., Skinner, 1938; Thorndike, 1927; Woodworth, 1899). In examining this association, researchers employed either relatively simple repetitive movements or reflexes to test their assumptions. For example, to examine the roles of memory and feedback during learning, Woodworth (1899) required blindfolded participants to draw lines of varying lengths. This research demonstrated that repetition of the movement without feedback did not improve accuracy during practice. However, when researchers provided simple right or wrong statements after each trial, participants' accuracy dramatically improved. These findings indicated the important role that feedback plays in reinforcing learning.

Other early researchers on skill acquisition believed that reflexes were the basic building blocks of movement and that people were passive recipients of external sensory inputs that drove their behavior. For example, to study movement in isolation of inputs from higher cortical brain regions, Sherrington (1906) conducted experiments on animals whose spinal cords had been surgically cut in order to study the circuitry of the central nervous system (CNS). Prominent experimental psychologists such as Skinner (1938) and Thorndike (1927) were also influential during this period. However, many researchers criticized these methods for assuming that animal models were relevant for studying motor skill acquisition in humans. Further, the research tended to focus too much on observable performance outcomes rather than

understanding of the underlying processes of skill acquisition. In addition, association theories typically lacked explanations to account for the fact that humans can choose when and how to moderate their actions, and they failed to explain that sensory feedback was not always essential for executing some movements, as in the case of rapid reactive actions.

Neuromaturational Theories

During the 1920s through 1940s, research on motor development eclipsed the study of skill acquisition. To some extent, there is no clear distinction between the disciplines of motor development and motor learning because both processes produce durable changes in motor behavior, although along different timescales. In the case of **motor development**, behavioral changes are typically attributed to growth and maturation, whereas in **motor learning**, practice or experience is the determining factor (Haywood & Getchell, 2005). Early researchers on motor development tended to explain behavioral change from a neuromaturational perspective, focusing particularly on the period from infancy through adolescence (Gesell, 1928; McGraw, 1943). This theory holds that the genetic predispositions that we inherit within our DNA determine the orderly sequence of each person's motor development (ontogeny). Many researchers viewed the stable progression of individuals through key motor milestones as persuasive evidence for the overriding influence of evolution and genetics over environment (see figure 1.2). Theorists suggested that the maturation of the CNS was the catalyst for the development of new movement skills and the gradual disappearance of older, less functional skills

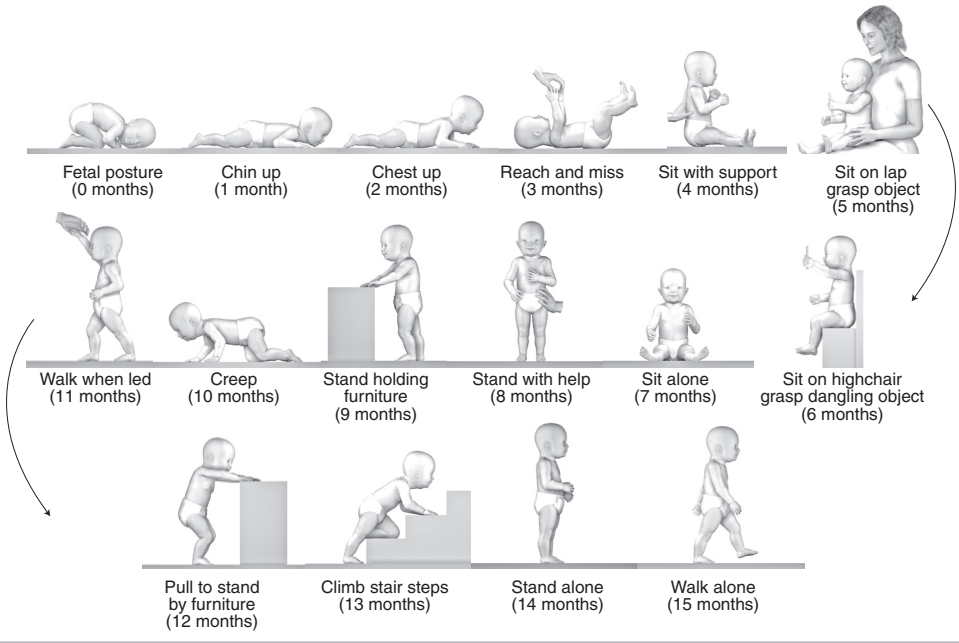


Figure 1.2 Milestones for motor development, infancy through adolescence.

such as early reflexes or motor babbling (considered by early researchers as random and incoherent limb movements of young infants).

However, in the absence of any direct measures of such maturation processes in the CNS, research in the field entered a period of descriptive analysis and identification of age-group norms from the mid-1940s to the 1970s. Researchers have subsequently focused more on interesting contradictions to the motor milestones progression (e.g., Thelen, Ridley-Johnson, & Fisher, 1983), which highlight the influence of each person's unique environment during development. Researchers no longer view individual variations in progression rates through motor milestones as potential evidence of pathologies. Instead, they believe that variations reflect the unique interactions between personal characteristics and the specific environmental context in which each person develops (Thelen & Smith, 1994).

Fitts' Stage Theory of Motor Learning

The second World War revived interest in processes of skill acquisition as governments in the United States and United Kingdom asked researchers to study methods for selecting and training military personnel. Due to pragmatic concerns, such as the ergonomic design of equipment and training of military skills, researchers adopted an overriding task-oriented approach that did little to shed light on processes of skill acquisition in humans (Schmidt & Lee, 2005). The emphasis on studying performance and learning in specific tasks resulted in knowledge on practical issues such as the optimal scheduling of practice and feedback delivery to enhance learning (e.g., amount, frequency, duration). Paul Fitts (1964) developed an influential model of skill acquisition that has received a great deal of support in subsequent textbooks and practical literature (e.g., Abernethy, 2001). His three-stage model describes learning as a continuous process with gradual changes in the nature of information processing as learning progresses (see figure 1.3).

During the initial **cognitive stage** of Fitts' model, the learner is exposed to simple rules and verbal instructions to acquire basic understanding of the movement. At this stage, performance is variable and error ridden as the learner experiments with different movement configurations. When the learner attains the following **associative stage**, movement patterns are refined and more consistent. Depending on the complexity of the task and the learner's abilities, this stage can require varying lengths of practice. Reaching the third **autonomous stage** requires extensive practice. At this stage the learner can perform the skill with minimal mental effort and few errors. As noted in the next chapter, much of the learning process inferred within Fitts' three-stage model has been attributed to the development and acquisition of information-processing abilities (Anderson, 1982) and the construction of relevant motor programs.

Information-Processing Theories

Beginning with the early writings of James (1890), Woodworth (1899), and Lashley (1917), a popular theory postulated that perceptual-motor informa-

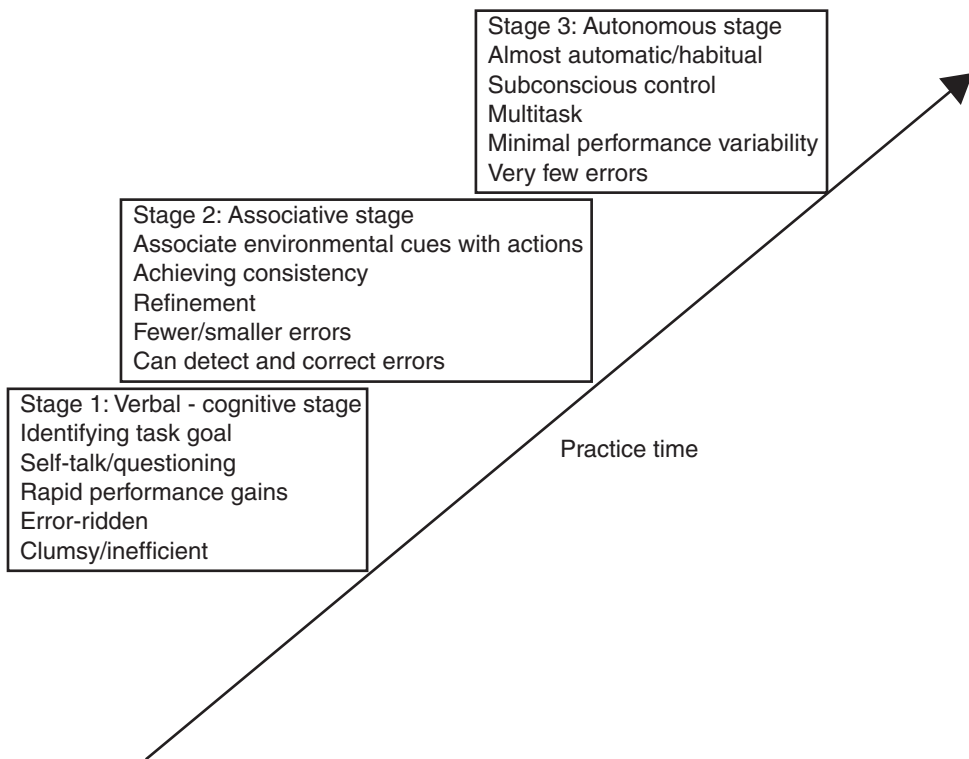


Figure 1.3 Fitts' three-stage model describes learning as a continuous process with gradual changes in the nature of information processing as learning progresses.

tion can be represented within the CNS. Such representations are acquired through learning, which results in the CNS storing a set of motor commands that control movement behavior. Researchers thought that this mode of movement regulation, known as **open-loop control**, was responsible for the control of quick, ballistic movements in which a set of preplanned instructions for action could be executed without feedback modification, for example in the performance of a boxing jab. During the years following World War II, Henry and Rogers (1960) emphasized the importance of centralized instructions for movement with their memory drum theory.

Traditionally, theories emphasizing the role of central structures in movement control have tended to look to electronic and mechanical storage devices to model how information can be stored and represented in the CNS. These devices have served as metaphors for many processes of movement control and demonstrate the influence of physics on psychological theories of movement behavior. For example, memory drums were devices in early computers onto which programs and information could be loaded for storage. Keele (1968) first used the term **motor program**, which he adopted for a popular functional analogy of how the brain produces consistent and reliable movement outputs, again mirroring the technological developments in computer hardware at the

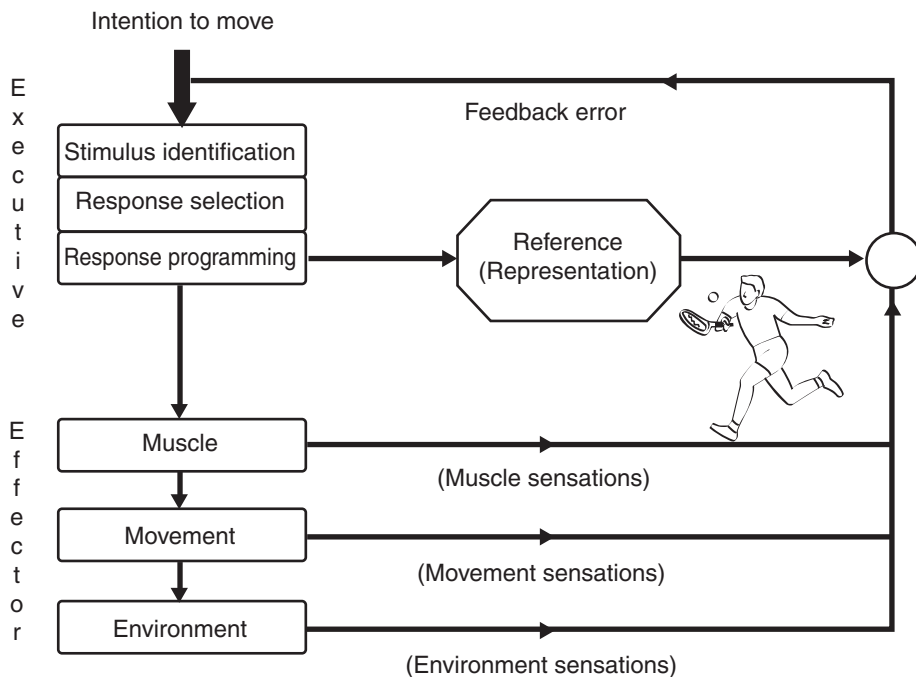


Figure 1.4 Information processing occurs in the CNS through a series of discrete cognitive stages involving perception, decision making, and response execution.

time (see Key Concept: Motor Program). The basic assumption of the analogy is that the brain functions like a computer to process information and produce outputs in behavior. Information processing in the CNS occurs through a series of discrete cognitive stages involving perception, decision making, and response execution (see figure 1.4). Researchers employed movement reaction time as a key variable to explain the stages of information-processing tasks that range in complexity (Keele & Posner, 1968; Klapp, 1977).



Key Concept

Motor program describes a set of movement commands that are stored within the CNS. Initially proposed by Keele (1968), the term draws heavily on the analogy of the brain as a sophisticated computer. The generalized motor program was subsequently developed to represent a general category of movements to deal with storage concerns (Schmidt, 1975). Critics of the motor program argue that the concept assumes the existence of a homunculus-type agent (in neurological sciences known as “the little man inside the brain”) or executive agent necessary to create and select the programs.

During the 1960s and 1970s, a major paradigm debate raged over the importance of the motor program versus the use of ongoing performance-related feedback to control movement. Adams' (1971) **closed-loop theory** advocated the role of feedback in both the control and learning of movement. Based on engineering principles in cybernetic science, closed-loop theory implies that normal movement is regulated by a continuous comparison between current sensory information and information generated as a consequence of a successful movement (see figure 1.5). Early learners develop representations (perceptual traces) of the expected movement by using outcome-related feedback (knowledge of results) to gauge their progress. Much published research supported Adams' theory, although it soon became apparent that, in contrast to open-loop theories, mainly slow positioning movements (e.g., steering a car or riding a bicycle) fit closed-loop theory best. In both open- and closed-loop theories, a key assumption is the capacity of the CNS to store a huge number of representations to control the many distinct movements needed in daily life. Just imagine the thousands of distinct movement patterns that humans might be called upon to perform in given circumstances.

Schmidt (1975) attempted to resolve the debate by combining aspects of open- and **closed-loop control** in proposing his **schema theory** of discrete motor skill learning (see Spotlight on Research). A **schema** is a set of rules concerning the execution of a movement response linked to feedback received during and after performance. In Schmidt's schema theory, a generalized motor program (GMP) is an abstract representation that contains the general

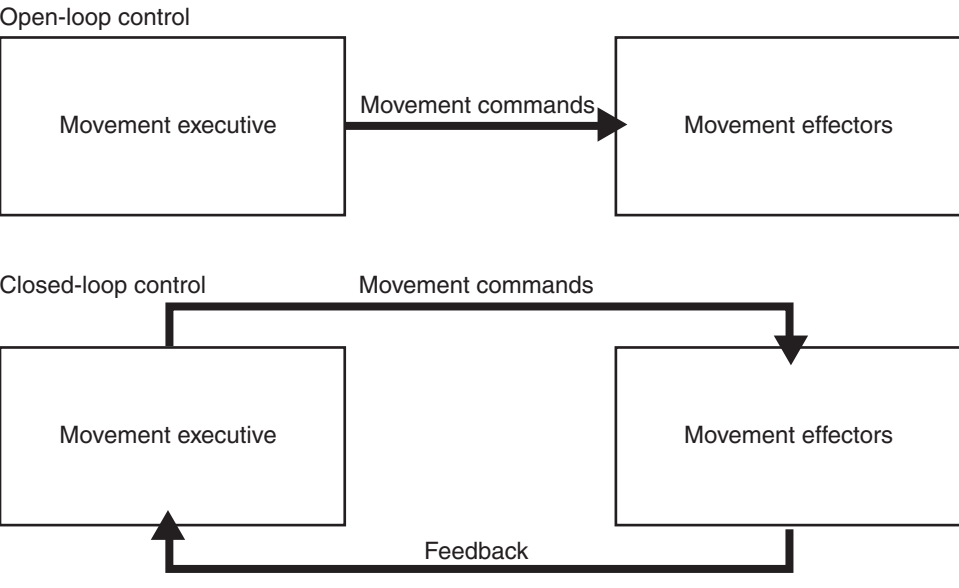


Figure 1.5 In closed-loop theory, early learners develop representations of the expected movement by using outcome-related feedback (knowledge of results) to gauge their progress.

characteristics for a given class of movements such as pointing, balancing, jumping, and locomotion. One implication of schema theory is that variable, rather than specific, practice conditions are necessary for the learner to acquire a robust schema that can generate functional movements for a particular class of actions (e.g., driving a car) under a variety of performance conditions. It is believed that driving a car under different environmental conditions will lead to a broader range of experience and allow drivers to cope with novel driving conditions (see Spotlight on Research). Schmidt's theory of learning provoked a good deal of laboratory research that tested aspects such as movement parameterization and variability of practice. Once more, however, there was a tendency to employ simple motor tasks involving few motor system components (e.g., limb segments) in restricted learning environments.



Spotlight on Research

SCHEMA THEORY OF DISCRETE MOTOR SKILL LEARNING

In 1975, Richard Schmidt published his original description of schema theory in the *Psychological Review*. Schmidt's concerns over the generalization of Adams' (1971) closed-loop theory beyond the learning of slow positioning movements motivated his publication. Subsequent research has cited schema theory frequently, with 782 citations by the end of 2004 (Shea & Wulf, 2005). Further, schema theory has contributed to a large body of empirical research across a broad range of performance domains, testing concepts such as relative timing and variability of practice.

Schmidt proposed the existence of GMPs to counter concerns that every movement would need a separate motor program, leading to storage problems. Instead, a given class of actions (e.g., locomotion, including stepping, walking, running, skipping, and gamboling) could be represented by the same GMP. To produce a response for a given situation, the learner needs to set key specifications or parameters, such as movement speed, duration, and force. Schema theory contends that with practice, learners should rely less on using feedback, instead developing open-loop control via parameter regulation to free this cognitive burden. To explain the gradual shift in control over learning, Schmidt identified two types of schema or abstract rules: recall schema and recognition schema. A schema is constructed from the different information sources that are stored each time a skill is produced, such as the initial conditions, the response specifications, the sensory consequences, and the response outcome. During practice, these information sources can be linked together to construct a recall and a recognition schema that the learner may apply even when performing in novel conditions. The following summarizes the attributes of schema theory:

- Schema theory explains how learners can produce specific movements of a given class that they had never performed previously.
- Variable practice conditions facilitate the creation of a robust, general schema that can lead to enhanced skill transfer.

- Separate memory structures are used for movement production (recall) and evaluation of the accuracy of selected movements (recognition).
- Error-detection capability will improve with practice.

Neurocomputational Theories

The 1990s were designated as the Decade of the Brain by the U.S. Congress, coinciding with the greater integration of neuroscientific evidence of CNS function and psychological modeling of behavior. This concerted integrative modeling effort was reflected in revisions of many motor control theories in the cognitive and behavioral neurosciences (e.g., Kawato, 1999). These internal models address early concerns of how movement representations might interact with rapid environmental changes to provide flexible movement solutions for individual performers. For example, Willingham (1998) proposed a **control-based learning theory (COBALT)** in which the brain creates different types of representations so that important information can be symbolically represented and stored.

According to COBALT, motor learning results in the acquisition of different types of representations in the brain based on features of the world, intentions, or sequential patterns of spatial locations or specific muscle forces (for a more detailed analysis, see Davids, Kingsbury, Bennett, & Handford, 2001). COBALT exemplified a neurocomputational perspective that proposed how representations could allow learners to create mental models of the world and include categories based on environmental features, intentions, and spatial or temporal patterns of muscle activation at different levels of the CNS. For example, a tennis serve is based on a learned egocentric representation (perceptions from the individual's viewpoint) of the "knowledge of a sequence of locations to which one should respond" (Willingham, 1998, p. 574). Arbib, Érdi, and Szentágothai (1998) proposed a similar use of hypothetical structures mediating brain function when they suggested that a schema is a unit of processing that intervenes between the neural structure and movement system dynamics.

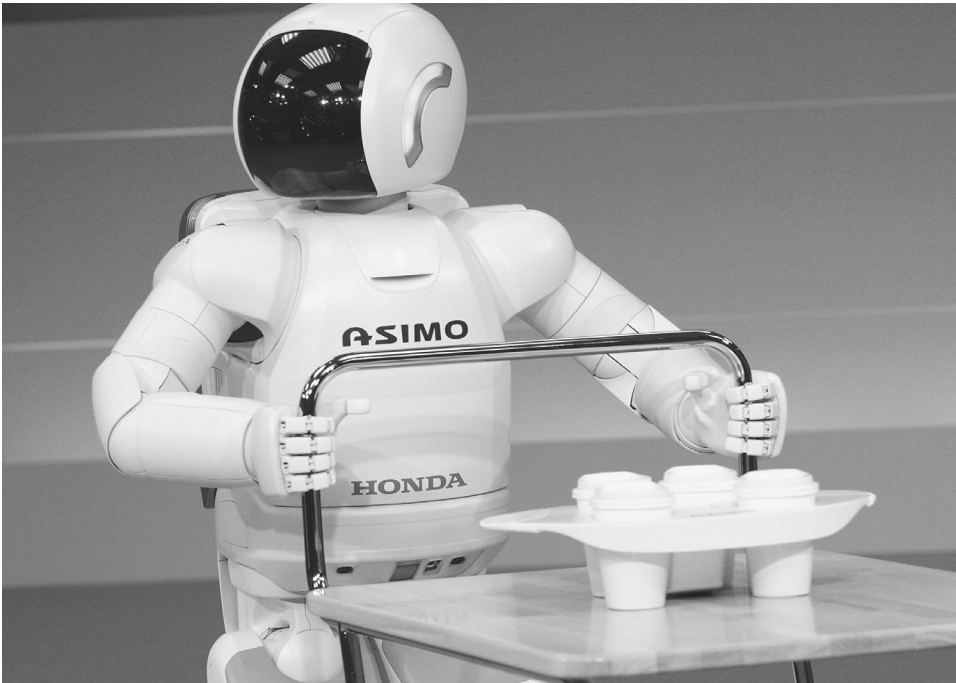
In neurocomputational approaches, researchers have been primarily concerned with identifying the regions of the brain that are responsible for controlling action. For example, Muellbacher and colleagues (2002) suggested that the primary motor cortex is the site of consolidation or stabilization of movement representations, particularly in the early stages of learning a new motor skill. Another common approach has been to consider how the brain uses its networked, parallel processing capabilities to cope with multiple information streams (e.g., through neural mechanisms such as cortical cell assemblies and neural net modeling) (Bullock & Grossberg, 1989; Wickens, Hyland, & Anson, 1994).

A major challenge facing computational models concerns the acquisition of coordination during complex multiarticular actions (actions involving many limbs, segments, and joints of the body) in changing environments. Another important question for computational theories concerns the communication among levels of representations, such as spatiotemporal representations of the environment and representations of movement patterns. Where and how does this communication occur in the CNS? More prosaically, how is the neural code transformed from egocentric representations of time and spatial locations into the CNS coding of specific muscle forces used to produce a movement pattern (Flash & Sejnowski, 2001)? In most early computational work, researchers hypothesized that interneurons (perhaps at the level of the spinal cord) acted as code translators for communicating between egocentric representations of the world and the specific muscles used to achieve movement goals.

Recent trends in computational modeling have been driven by attempts to design and engineer biologically inspired robots (i.e., robots that act more like common neurobiological systems, including insects, animals, and humans). This generation of robotic design is tackling the tricky problem of engineering machines that can complete challenging tasks, adapt their behaviors to complex environments and terrains, and even cooperate in teams when engaged in complex tasks like exploration for minerals during mining (e.g., Mataric, 1998). The result is modeling of artificial intelligence that, according to some researchers, provides insights into how the CNS might control movements in humans. Scientific subdisciplines such as robotics and artificial intelligence are now drawing inspiration from research on how neurobiological systems interact with complex environments. Traditional models of robotic design, based on the principles of sense, model, plan, and act, are being complemented by behavior-based control approaches that have the explicit goal of engineering machines to operate in unpredictable and challenging environments, such as during space exploration or RoboCup soccer competitions (Mataric, 1998). Alternative design principles, including situated robotics, draw inspiration from research on how neurobiological systems search for and pick up information to support sophisticated interactions with their environments (e.g., Di Paolo, 2002; Paine & Tani, 2005).

Some behavior-based robotic implementations have been founded on ecological principles in which an agent and its environment are viewed as an inseparable system studied at an ecological scale of analysis. In ecological robotics, adaptive movement behavior is specified by information in the environment that does not need to be represented within the robot in explicit detail (Fajen et al., 2003). There have even been attempts to forego a central planner for behaviors and to map sensory systems directly to actuators of the robots (Duchon & Warren, 2002).

The influence of neurobiological theories on robotic engineering has not been a one-way street. As a result, these newer models of artificial intelligence



Robots that can complete challenging tasks, adapt their behaviors to complex environments, and cooperate in teams help researchers determine how the CNS might control movements in humans.

have inspired an emphasis on understanding how the coadaptation of the CNS and the body occurs during motor learning (e.g., Davidson & Wolpert, 2003; Wolpert, Ghahramani, & Flanagan, 2001). This advance is significant because it recognizes how cognitive processes can be mediated by the physical constraints of the human body, which changes over the life span as a result of development and over generations as a result of evolutionary pressures.

In this type of computational modeling, anticipating the outcomes of an action forms the basis for planning and organizing goal-directed movement. Communication between sensory representations and motor systems is resolved by requiring two types of representations: forward and inverse models (Kawato, 1999). **Forward modeling** refers to the mapping of motor commands to the related sensory consequences of a movement. Located within the brain, this representation is used to predict the sensory consequences of planned movements. For example, when planning the action of reaching for an object on a high shelf, expected sensory consequences might include information from leaning and overbalancing that would require adaptation of postural muscles and possibly even a stepping adjustment to facilitate the elongated reach to the shelf.

Inverse modeling refers to transforming the anticipated sensory consequences into the motor commands needed to achieve them. From this

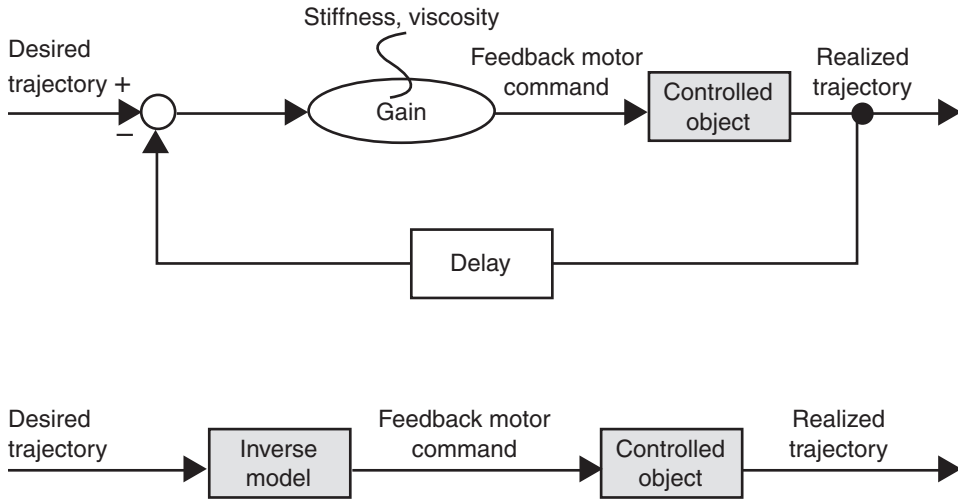


Figure 1.6 With forward models, one learns to use feedback to regulate movements. With task constraints that are too rapid to process information, inverse models provide performers with feed-forward control.

perspective, motor learning involves the acquisition of forward and inverse representations for varying tasks and environments (see figure 1.6). With forward models, one learns to use feedback to regulate movements, whereas under task constraints, which are too rapid to process this information, inverse models provide performers with feed-forward control. The acquisition of forward models enables people to adapt to rapidly changing environments, prevents them from becoming too reliant on immediate sensory information to regulate movements, and possibly forms the basis of using imagery to mentally practice a skill (Wolpert et al., 2001).

As long as environmental perturbations requiring adaptation remain within the performers' experience, feed-forward control allows performers to use 'what-if' mental simulations in a kind of preplanned reactive adaptation to a change in the environment. For example, experienced tennis players might predict that an opponent's forehand drive, played with a very low arc at around net height, could just clip the top of the net, changing the flight trajectory of the ball. This possible consequence might lead to feed-forward control of a pickup volley action as one option in those circumstances. Regulated by inverse models, the neural mechanism of feed-forward control recognizes that biological feedback loops act quite slowly and overcomes delays in controlling very rapid or ballistic movements through sensory information. In these neuro-computational models, complexity and storage capacity issues are addressed by the proposal that the CNS actively condenses the detail and complexity of the information it stores by reducing the dimensionality of the forward and inverse representations.

Common Features of Traditional Theories

Traditional theories of motor skill acquisition share several key characteristics. These features are useful to consider when comparing traditional theories with the alternative theories described in subsequent chapters. These commonalities include the role of representations in controlling perception and action, a preference for hierarchical control modeling, information storage within neurobiological systems, noise viewed as system error, and a preference for studying specific types of movements, such as manual aiming, pointing, and reaching and grasping. In this section, we highlight these common features.

Representations of the World and Movements

Traditional theories have been inspired by advances in cognitive science, which tend to compare brain structures to computer architecture and mental processes to computer algorithms. As noted earlier, this theoretical rationale for motor skill acquisition and control uses the base metaphor of an internal representation of the world to describe how perception, cognition and action can occur. This metaphor demonstrates the influence of physics on psychological science where the dominant research agenda over the past five decades invoked a view of intentions as representations which function as internal causes of human behavior (van Orden et al., 2003). In psychology, representations have been a popular metaphor for cause-and-effect relations between intentions and movement behavior. This metaphor has not only exerted a strong influence on the design of psychology experiments but also the interpretation and analysis of data and the theoretical explanations preferred by psychologists (van Orden et al., 2003).

Accordingly, these theories infer that learning involves constructing internal models of the world and of movements that facilitate interactions with the environment. Researchers believe that these representations support the internalization of information from the environment that can be used to guide actions. Early representational theories were plagued by problems of specificity, storage capacity, and computational complexity, but recent work has been informed by developments in computational neural network theory, neuroscience, artificial intelligence, and robotics to model how people can acquire adaptive and reliable movements. Developments in this area are taking advantage of increased understanding of the CNS to constrain modeling of representational control of behavior.

Learner as Hierarchical Control System

Traditional theories assume that the coordination and control of movement requires some kind of ordered prescription or representation for action (Anson, Elliott, & Davids, 2005; Newell, 2003). Traditional cognitive approaches deal

with this requirement by invoking a central controlling mechanism that resides in the brain. Often known as the **homunculus**, this mechanism is responsible for the creation, selection, and timing of motor programs. Hierarchical control systems operate in a top-down fashion and are analogous to a puppeteer who controls each movement of a puppet by controlling the movement of individual strings attached to the parts of the puppet. Another example of hierarchical control systems exists in the social organization of companies in which all decisions are made by the manager, whose instructions to lower-level supervisors ultimately result in the outputs of workers. In such systems, the emphasis for skill acquisition relies heavily on constructing increasingly elaborate cognitive processes to coordinate increasingly skillful movement patterns. Additionally, hierarchical systems largely ignore a learner's previ-

ous experience and the physical makeup of the movement system; feedback processes and the specific properties of each person's body are underplayed (see chapter 2).

Capacity-Limited Storage

Borrowing a metaphor from computer science, movements are regulated by symbol-based programs that must be stored within the CNS so that they can be recalled for action when necessary. Despite significant progress in the fields of neurobiology and neuropsychology, how and where motor programs are coded and stored remains uncertain. A further issue arising from cognitive approaches is that, like a computer, the function of the CNS is capacity limited. Researchers have cited this limitation to explain why the allocation of attention is such an important characteristic of skilled behavior (Abernethy, Maxwell, Masters, van der Kamp, & Jackson, 2006).



Hierarchical control systems regard an executive or external agent as responsible for system organization, as analogized by the puppeteer controlling the moving parts of a puppet.

Indirect Perception

In order to make sense of the world in which we move, traditional theories assume that the performer needs to interpret information internally before responding to it. In other words, meaningless sensory information must be compared with existing representations of the world (memories) that are stored in the brain. For novice learners, such indirect processing would be burdensome as they try to make sense of their environment without previous experience upon which to rely. Conversely, skilled performers have learned to identify key information sources quickly and use their existing knowledge to alleviate the processing burden. For example, experienced mountaineers can look at the face of a rock and detect the most useful hand- and footholds on the surface. Their expertise allows them to perceive subtle information sources concerning size, depth, and width of surface characteristics that less experienced mountaineers do not perceive. Moreover, their speed of perception is enhanced by the strategy of chunking information into larger, more manageable wholes. They do not need to perceive and identify smaller units of information in a time-consuming process.

Reduction of Error and Noise

Traditionally, cognitive science has viewed high levels of movement variability as a problem for humans during task performance (Newell, Deutsch, Sosnoff, & Mayer-Kress, 2006). For example, the information-processing approach views the performer as a sort of human communications channel in which the relationship between input signals and system output is linear and deterministic. In this view, the performer can eliminate or minimize noise through practice and task experience. For this reason, the magnitude of variability in performance has been viewed as an important feature for assessing the quality of system control (Schmidt, 1985). The role of repetitive practice is often conceived of as gradually reducing the amount of movement pattern variability viewed as noise. But is movement variability always bad? As we shall learn later in this chapter, these theoretical approaches have typically neglected the important structural role that variability might play in adapting successful goal-directed movement patterns to changing environments.

Models of Limited Range of Movement

In the field of human movement, a key task is to understand how the movement system achieves coordination among components of the body and among movement patterns and important objects and surfaces in the environment. That is, humans need to acquire skill in coordinating relevant parts of the body into a functional pattern identifiable as a coherent action, such as when performing a triple salko jump in ice-skating, or more mundanely, when stepping onto a surface. However, in both of these examples, even the most coherent

movement pattern needs to be coordinated with respect to an environmental surface or obstacle, for example when landing on ice or when placing the foot on a step or curb. Because of the potential for interaction among these subsystems, coordination in the human movement system is characterized by great complexity (Davids, Bennett, & Newell, 2006).

The search for simplistic laws of cause and effect has meant that movement scientists have often struggled to come to terms with the complexity humans must overcome to produce even a simple goal-directed movement. This issue has become known as Bernstein's (1967) degrees of freedom problem.

Degrees of freedom (dfs) are the independent components of a complex system that can fit together in many different ways. For example, insects in a colony are independent dfs whose actions need to be coordinated to build a nest or feed progeny. In the same way, the large number of available mus-

cles, joints, limb segments, and bones in the body exemplify the dfs of a complex system. This huge number of dfs in the human motor system is a kind of a curse for computational accounts of skill acquisition and performance (Kugler & Turvey, 1987). This is because it seems that a hierarchical control system with an executive agent in charge needs to be aware of so much information on where each individual part is in space, what it is currently doing, and how its position in space might change over time during performance.

Traditionally, the selection of movement models to investigate motor system functioning has been biased away from dynamic, multijoint, functional actions prevalent in sports because of the view that experimental rigor can be better maintained in laboratory studies of simple movements (Davids, Button, Renshaw,



Complex systems in nature are composed of many parts that are free to interact with each other. Many examples abound in nature, as exemplified by cloud formations.

Araújo, & Hristovski, 2006). Consequently, many traditional theories were conceived to explain a small group of related actions (Newell, 1989). For example, open-loop theories are best suited to fast, ballistic movements; closed-loop theory deals with slow, precise movements; and schema theory was devised to explain the learning of discrete movements. Traditional approaches to skill acquisition have not offered a unified theory of learning (Anson et al., 2005).

Historically, there has been an overemphasis on the amount of change in performance outcomes without sufficient analysis of the dynamic properties of change in movement coordination (Newell, 2003). This criticism may be partly attributed to the lack of sophisticated measuring devices for examining coordination (e.g., three-dimensional motion analysis systems). Another concern is that much skill acquisition research has only been conducted over short, intense practice periods (e.g., a number of days). There are few examples in the research literature of motor learning experiments performed over longer periods of practice, as experienced in real life (e.g., the months and years of practice involved in many work and sport environments), and conclusions are likely to be based on the relatively transient effects of practice rather than the permanent consequences of learning.

Implications for Movement Practitioners

Surprisingly, the links between skill acquisition theories and practice have not always been exploited to the extent that one might expect (Fairweather, 1999; Newell & Rovegno, 1991). Nevertheless, many movement practitioners such as therapists, teachers, and coaches have implicitly adopted the learner model developed by traditional theories. In this section, we will discuss some of the key implications behind translating accepted theoretical approaches into practice.

Amount of Practice

Because movement practitioners often have a limited amount of time to work with learners, it is interesting to consider how much practice learners require in order to achieve different levels of expertise. Unfortunately, due to practical difficulties of recruiting participants and monitoring progress over time, researchers have conducted very few long-term studies of motor learning. Notable exceptions include Snoddy (1926) and Crossman (1959). Analysis of expert performers' practice activities are perhaps the most illuminating. Some studies have reported that a minimum of 10 years of practice is necessary to reach exceptional performance levels. Ericsson, Krampe, and Tesch-Römer (1993), who studied musicians, suggested that

cumulative hours spent engaging in deliberate practice were monotonically and positively related to performance level. It was argued that, within certain physical limits, the more hours you spend in deliberate practice, the more likely you are to develop a high level of expertise in a particular performance context. In terms of negotiating Fitts' (1964) three stages of learning, researchers have remained silent on the required practice duration. In summary, the traditional take-home message for practice seems to be that more is best as researchers have concentrated on studying the amount of time needed to acquire a high level of expertise.

Practice Variability

Consider an amputee learning to walk with a new prosthesis or a new 10-pin bowler trying to perform a strike. Do such learners benefit from variations in practice conditions, or should they concentrate on simple repetitions designed to produce a constant movement pattern? The answer largely depends on how much variability in conditions the learner might experience in performance situations. In stable, predictable conditions such as in bowling, traditional approaches would recommend a good deal of trial-to-trial repetition of relevant skills such as foot placement, ball release, and follow-through to consolidate the motor program responsible for that action. For dynamic tasks such as negotiating a cluttered environment or a busy street, the amputee would need to practice walking in different conditions to feel comfortable with a prosthesis. As discussed in this chapter's Spotlight on Research, Schmidt's schema theory (1975) recommends practice variations in the key parameters that influence the production of GMPs, such as movement amplitude, force, or time. In a later article, Schmidt (1985) pointed out that researchers and practitioners should seek practice activities that preserve the invariant aspects of a learner's coordination (e.g., relative timing).



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The results of research that consider variability of practice have been quite equivocal (see van Rossum, 1990). It seems that a number of factors mediate the potential benefits of variable practice conditions. For example, younger, inexperienced learners seem to benefit more from variable practice conditions than older learners (possibly suggesting how discovery learning may benefit that group). On the whole, the learning of ballistic and complex tasks also seems to respond well to variable practice (for a review, see Lee, Chamberlin, & Hodges, 2001). Because movements cannot be stereotypically repeated in an identical manner (Newell et al., 2006), the role of repetition in practice has come under scrutiny. It seems that requiring learners to adapt their movements under variable task constraints has considerable merit in helping them learn how to interact with complex environments.

Practice Organization

Firefighters and people who fish need to learn how to tie different types of knots for quite different purposes in their working environments. Should they learn to tie these knots in a blocked fashion (one knot at a time), or should they practice different knots within the same practice session (randomized practice)? Researchers' experiments in the area of practice organization have led to a well-known research phenomenon called the **contextual interference effect** (see Brady, 1998). Practice organization that is blocked seems to lead to better performance during acquisition of the skill; however, random practice organization benefits the learning (i.e., retention and transfer) of a skill. This effect seems to be most potent when different GMPs are randomly assorted during the practice (Schmidt, 2003). In addition, although the contextual interference effect has received considerable support in the literature, modifying variables such as skill complexity and skill level need to be taken into account (Ollis, Button, & Fairweather, 2005).

Surprisingly, adding random variability in the form of irrelevant movement components when practicing a target movement leads to better performance and learning than when practicing specific movement patterns alone (see Button, Davids, & Schöllhorn, 2006). This effect was evident in studies where soccer players practiced shooting at the goal with the addition of superfluous hopping movements or jumps (Schöllhorn et al., 2006). Traditional theorists argue that random practice seems to emphasize enhanced cognitive processing during practice, which in turn leads to more effective motor learning.

Part-Task Practice

Swimming instructors often ask their students to perform different elements of a stroke in isolation (e.g., leg kicks, arm pulls, and breathing) before fitting them all together to produce the whole pattern at a later stage of practice. Is this strategy consistent with theoretical modeling? Performing complex motor



Swimming instructors often ask their students to perform different elements of a stroke in isolation before fitting them all together at a later stage.

skills represents a considerable information-processing burden, particularly for early learners (Wulf & Shea, 2002), so movement practitioners have been advised to seek strategies that reduce the cognitive load, at least early in learning. In the swimming example, the instructor's implicit assumption is that learning one part of the task, such as the leg kick, will transfer positively to learning the whole task.

However, research has revealed once more that the nature of the task influences the value of this common practical strategy. In particular, tasks that have highly interdependent parts or complex coordination requirements (e.g., snowboarding jumps) do not always benefit greatly from part-task practice. Part-task practice can take different forms, such as segmentation, where sequences of movements are practiced separately, or fractionation, as in the swimming example. In general, due to an individual's potential to learn chunks of a movement more effectively, traditional theories support the concept of part-task practice.

Instructions and Feedback

Ask any teacher about the importance of giving prepractice instructions to students, and it is likely you will receive an enthusiastic response. From a traditional perspective, instructions facilitate the skill acquisition process by specifying an optimal movement pattern to the learner, thereby strengthening the representation of the desired action. Verbal feedback also is thought to be particularly useful because it helps learners understand when or why they made mistakes. The frequency of verbal input should gradually decrease over time so that learners can begin to rely on their own sensory systems to gauge their progress.

It is difficult to determine from past research how detailed instructions need to be to optimize skill acquisition. Once more, the information-processing approach suggests that early learners benefit from a limited number of simple instructions, whereas advanced learners are better equipped to deal with more detailed instructions. Practitioners also can use instructions to help focus the learner's attention on certain aspects of the task. In this regard, Wulf and colleagues (e.g., Wulf, McConnel, Gartner, & Schwarz, 2002; Wulf, McNevin & Shea, 2001) recommend using external-focus instructions that direct learners to the external consequences of their movement patterns rather than the internal processes of producing the required movement. For example, when learning to ride a bicycle, an external focus of attention might be the orientation and direction of the front wheel as the rider moves through the environment, whereas an internal focus might be what the legs and hands are doing with the pedals and handlebars. Instructions delivered as movement analogies have also been shown to be positive in reducing the learner's processing load (Liao & Masters, 2001).

Demonstrations of Technique

Providing a visual demonstration of the required technique is another way to present augmented information to the learner (Ashford et al., 2006). Many practitioners believe that a demonstration provides a visual model that the learner may more easily interpret than verbal commands. Further, many researchers encourage practitioners to use demonstrations early in learning to assist the coding of new information into a cognitive representation of a skill (Bandura, 1977). Although demonstrations have been shown to promote task consistency during learning, there is some concern that reproducing a particular movement pattern should not be viewed as achieving the task goal (Lee et al., 2001). In addition, there is still uncertainty over whether learners benefit from watching other learners or whether practitioners themselves should provide the demonstration (Magill, 2006). Horn and Williams (2004) suggested that the nature of the task dictates prescriptive advice regarding modeling (see table 1.1). For example, the learning of outcome-defined tasks, such as repairing a car engine or cooking a meal, benefits from expert models alternated frequently with practice, whereas process-defined tasks, such as a

TABLE 1.1 *Research-Based Prescriptive Advice*

Increasingly outcome driven ← Increasingly process-driven →	Type of task	Description	Examples	Advice	Research support
	Outcome-defined	The outcome of the action defines performance. The process used to achieve the outcome is irrelevant	Computer tracking and game tasks, mechanical assembly, perceptual anticipation	Use multiple demonstrations for perceptual anticipation Make the strategy salient Alternate demonstration with practice Emphasize the action of objects rather than the model.	Weeks (1992); Weeks and Choi (1992) Burwitz (1975) Blandin et al. (1994) Hodges and Franks (2002)
	Outcome-dominated	The process or technique contributes to achieving the outcome, but outcome dominates the assessment of task performance	Higher cognitive: Semaphore and manual alphabet, movement sequence activities. Lower cognitive: Weight lifting; performing complex motor skills for successful outcomes	Demonstrations before practice Encourage mental rehearsal Increase cognitive engagement by alternating demo and practice Lower cognitive Include learning models Provide model's KR	Bandura and Jeffrey (1973) Weeks et al. (1996) McCullagh and Caird (1990) Herbert and Landin (1994)
	Process-dominated	Outcomes contribute, but process dominates the assessment of performance outcomes	Performing complex actions such as kicking, throwing, or hitting with outcomes, but emphasizing technique	Minimize early intrinsic KR to induce rapid improvements in form Use demonstration retrospectively as a reference of correctness	Horn et al. (2002 and in press) Richardson and Lee (199)
	Process-defined	The process defines performance Technique is the measured outcome, or no outcome is included	Diving, trampolining, gymnastics Performing techniques such as kicking, throwing, or hitting in absence of object or target	Guide learners' attention to key elements of the demonstration with verbal guidance Use slow motion to enhance relative motion	Champenoy and Janelle (1999) Scully and Carnegie (1998); Fehres and Olivier (1986)

Horn, R., & Williams, A.M. (2004). Observational learning: Is it time we took another look? In A.M. Williams & N. Hodges (Eds.), *Skill acquisition in sport: Research, theory and practice* (pp. 175-206). Abingdon, UK: Routledge, p. 198.

gymnastics routine, may require slow-motion demonstrations combined with verbal guidance.

Summary

The scientific study of natural phenomena has always been influenced by the sociological and cultural influences of the time (Newell & Slifkin, 1998). In the study of human movement, theorists have been heavily influenced by the prevalent technology of the latter half of the 20th century, the so-called electronic and digital ages. Traditionally, information-processing approaches to motor learning and control have dominated sport science, kinesiology, and motor behavior, emphasizing the development of representations and plans for movement production (Rose & Christina, 2005). From this approach, the main goal has been to study the human brain, conceptualized as a capacity-limited, computational device into which raw sensory information is channeled and then processed with the help of stored representations of the world (e.g., Adams, 1971; Schmidt, 1975).

Traditional theories view symbolic movement plans as the major influence on how we control and learn actions. The increased efficiency and effectiveness of our movement representations help us to optimize and adaptively control our movements. A goal of cognitive neuroscience research has been to locate representations within the CNS, and practice is the process by which we increase the sophistication of these CNS-stored representations of the world and of our movements. Practitioners and physical therapists in many training and rehabilitation programs have implicitly adopted this popular view of the human performer. The traditional approach to motor learning and control has led to many advances in knowledge and has been described and evaluated in several important texts such as Schmidt and Lee (2005) and Magill (2006).

SELF-TEST QUESTIONS

1. Name some of the key research contributors to the development of skill acquisition theories. What features of skill acquisition did their work emphasize?
2. What common features do traditional models of skill acquisition share?
3. Apply Schmidt's schema theory to a movement skill of your choice. How should the skill be taught to a novice learner to optimize skill acquisition?
4. Compile a list of recommendations to be given to a movement practitioner. What considerations should be made concerning the task, the learner, and the practice environment?

ADDITIONAL READING

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